Algebraic Geometry

Shing Tak Lam

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Preliminaries

Plan

Roughly speaking, there are four parts of the course:

1. Spectrum of a commutative ring, and basics of sheaf theory.
2. Definitions of schemes and morphisms between them.
3. Properties of schemes, and properties of morphisms (replacements of naïve topological notions of compactness, Hausdorff, etc.). Basic constructions (Picard group, class group).
4. Rapid introduction to sheaf cohomology.

Prerequisites

(i) Basic undegraduate mathematics (groups, rings, topological spaces and so on).

(ii) Commutative algebra (co-requisite).

Resources

(i) Dhruv's course page.

(ii) Textbooks: Hartshorne, Vakil for Algebraic geometry, Atiyah-MacDonald for commutative algebra.

Why suffer through this? (Weil conjectures)

The Weil conjectures link two worlds. The starting data is a homogeneous polynomial $f \in \mathbb{Z}[x_0, \dots, x_n] = \mathbb{Z}[\mathbf{x}]$, of degree d.

World 1: We can consider the vanishing locus

$$X = \mathbb{V}(f) = \{(a_0 : \cdots : a_n) \mid f(a_0, \dots, a_n) = 0\} \subseteq \mathbb{P}^1_{\mathbb{C}}$$

which is a projective hypersurface. We will assume that X is smooth, i.e. there is no point $p \in X$, such that all of the partial derivatives $\frac{\partial f}{\partial x_i}$ vanish for all i. If we view $\mathbb{P}^n$ as $(\mathbb{C}^{n+1} \setminus 0)/\text{scaling}$, and $\mathbb{C}^{n+1} \setminus 0$ is given the Euclidean topology, then X is a (complex) manifold. This follows from the inverse/implicit function theorem. Moreover, X is a compact topological space. In fact, X can be triangulated.

With this, we can consider various invariants associated to X. The simplest is the Euler characteristic $\chi(X)$. For example, $\chi(\mathbb{P}^k) = k + 1$. More generally, we can consider the Betti numbers b_i, where

$$b_i = \dim_{\mathbb{Q}} H_i(X; \mathbb{Q})$$

are the dimension of the rational homology groups. In this case,

$$\chi(X) = \sum_i (-1)^i b_i$$

World 2: Fix a prime p, such that X is smooth over $\overline{\mathbb{F}_p}$.